Linear Regression

Libraries

The `library()` function is used to load *libraries*, or groups of functions and data sets that are not included in the base `R` distribution. Basic functions that perform least squares linear regression and other simple analyses come standard with the base distribution, but more exotic functions require additional libraries.

Here we load the 'faraway' and `MASS` packages, which are a very large collection of data sets and functions. We also load the `ISLR2` package, which includes the data sets we will use.

```
#install faraway package
install.packages("faraway")
```

```
## Installing package into 'C:/Users/hakan/AppData/Local/R/win-library/4.4'
## (as 'lib' is unspecified)

## Error in contrib.url(repos, "source"): trying to use CRAN without setting a mirror
```

```
#once package is installed, you don't need to run the code in line 18
#you need to load the package every time you start the R session
library(faraway)
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.4.3
```

```
install.packages("ISLR2")
```

```
## Installing package into 'C:/Users/hakan/AppData/Local/R/win-library/4.4'
## (as 'lib' is unspecified)

## Error in contrib.url(repos, "source"): trying to use CRAN without setting a mirror
```

```
library(ISLR2)
```

```
## Warning: package 'ISLR2' was built under R version 4.4.3

##
## Attaching package: 'ISLR2'

## The following object is masked from 'package:MASS':
##
##     Boston
```

If you receive an error message when loading any of these libraries, it likely indicates that the corresponding library has not yet been installed on your system. Some libraries, such as `MASS`, come with `R` and do not need to be separately installed on your computer. However, other packages, such as `ISLR2`, must be downloaded the first time they are used. This can be done directly from within `R`. For example, on a Windows system, select the `Install package` option under the `Packages` tab. After you select any mirror site, a list of available packages will appear. Simply select the package you wish to install and `R` will automatically download the package. Alternatively, this can be done at the `R` command line via `install.packages("ISLR2")`. This installation only needs to be done the first time you use a package. However, the `library()` function must be called within each `R` session.

Simple Linear Regression

The `ISLR2` library contains the `Boston` data set, which records `medv` (median house value) for 506 census tracts in Boston. We will seek to predict `medv` using 12 predictors such as `rmvar` (average number of rooms per house), `age` (proportion of owner-occupied units built prior to 1940) and `lstat` (percent of households with low socioeconomic status).

What is our dependent variable?

What are our independent variables?

```
#head comment shows the first six observation of the data
head(Boston)
```

```
##      crim zn indus chas   nox    rm  age    dis rad tax ptratio lstat medv
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3  4.98 24.0
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8  9.14 21.6
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8  4.03 34.7
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7  2.94 33.4
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7  5.33 36.2
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7  5.21 28.7
```

```
#following two lines shows the data dictionary
help(Boston)
```

```
## starting httpd help server ... done
```

```
?Boston
#str comment shows the data frame
str(Boston)
```

```
## 'data.frame':    506 obs. of  13 variables:
##  $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
##  $ zn     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
##  $ chas   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
##  $ rm     : num  6.58 6.42 7.18 7 7.15 ...
##  $ age    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ dis    : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ rad    : int  1 2 2 3 3 3 5 5 5 5 ...
##  $ tax    : num  296 242 242 222 222 222 311 311 311 311 ...
##  $ ptratio: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
##  $ lstat  : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ medv   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

We will start by using the `lm()` function to fit a simple linear regression model, with `medv` as the response and `lstat` as the predictor. The basic syntax is `lm(y ~ x, data)`, where `y` is the response, `x` is the predictor, and `data` is the data set in which these two variables are kept.

```
lm.fit = lm(medv ~ lstat,data=Boston)
```

The command causes an error because R does not know where to find the variables medv and lstat. The next line tells R that the variables are in Boston. If we attach Boston, the first line works fine because R now recognizes the variables.

```
lm.fit <- lm(medv ~ lstat, data = Boston)
attach(Boston)
lm.fit <- lm(medv ~ lstat)
```

If we type lm.fit, some basic information about the model is output. For more detailed information, we use summary(lm.fit). This gives us p-values and standard errors for the coefficients, as well as the R^2 statistic and F-statistic for the model.

```
lm.fit
```

```
##
## Call:
## lm(formula = medv ~ lstat)
##
## Coefficients:
## (Intercept)        lstat
##       34.55        -0.95
```

```
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.168  -3.990  -1.318   2.034  24.500
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 34.55384    0.56263   61.41   <2e-16 ***
## lstat       -0.95005    0.03873  -24.53   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.216 on 504 degrees of freedom
## Multiple R-squared:  0.5441, Adjusted R-squared:  0.5432
## F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

We can use the names() function in order to find out what other pieces of information are stored in lm.fit. Although we can extract these quantities by name—e.g. lm.fit$coefficients—it is safer to use the extractor functions like coef() to access them.

```
names(lm.fit)
```

```
##  [1] "coefficients"  "residuals"     "effects"       "rank"
##  [5] "fitted.values" "assign"        "qr"            "df.residual"
##  [9] "xlevels"       "call"          "terms"         "model"
```

```
coef(lm.fit)
```

```
## (Intercept)       lstat
##  34.5538409  -0.9500494
```

In order to obtain a confidence interval for the coefficient estimates, we can use the `confint()` command.

```
confint(lm.fit)
```

```
##                 2.5 %     97.5 %
## (Intercept) 33.448457 35.6592247
## lstat       -1.026148 -0.8739505
```

The `predict()` function can be used to produce confidence intervals and prediction intervals for the prediction of `medv` for a given value of `lstat`. We will talk about the prediction and confidence intervals later in the class.

```
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))),
    interval = "confidence")
```

```
##        fit      lwr      upr
## 1 29.80359 29.00741 30.59978
## 2 25.05335 24.47413 25.63256
## 3 20.30310 19.73159 20.87461
```

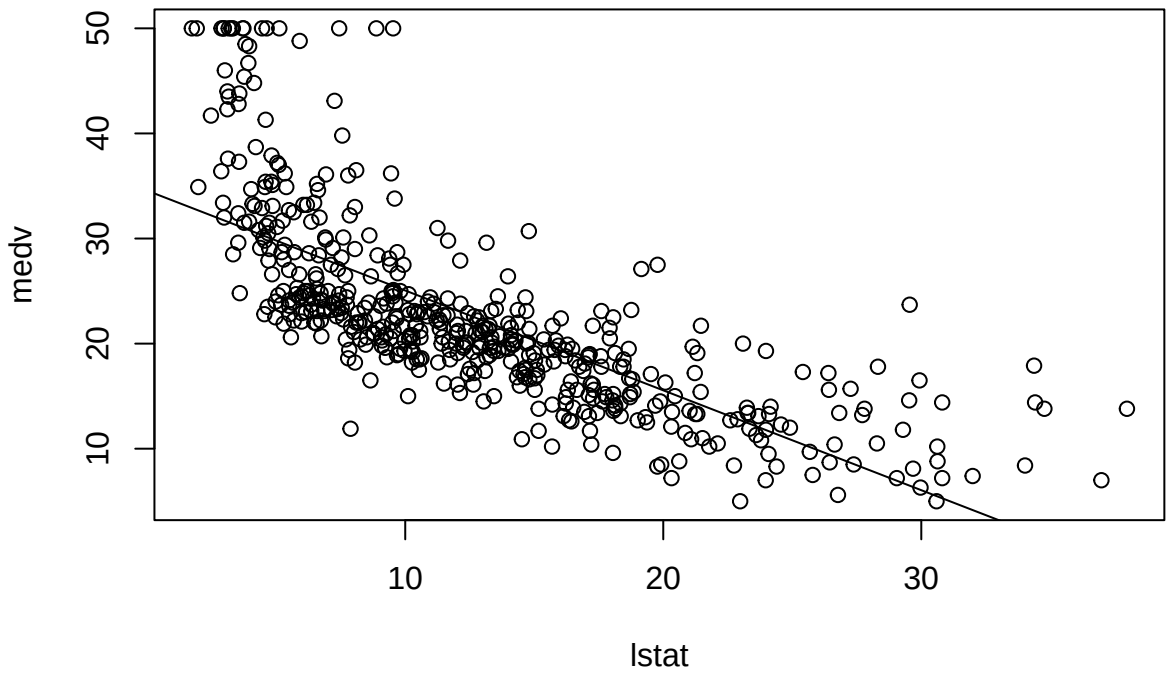
```
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))),
    interval = "prediction")
```

```
##        fit       lwr      upr
## 1 29.80359 17.565675 42.04151
## 2 25.05335 12.827626 37.27907
## 3 20.30310  8.077742 32.52846
```

For instance, the 95 % confidence interval associated with a `lstat` value of 10 is $(24.47, 25.63)$, and the 95 % prediction interval is $(12.828, 37.28)$. As expected, the confidence and prediction intervals are centered around the same point (a predicted value of 25.05 for `medv` when `lstat` equals 10), but the latter are substantially wider.

We will now plot `medv` and `lstat` along with the least squares regression line using the `plot()` and `abline()` functions.

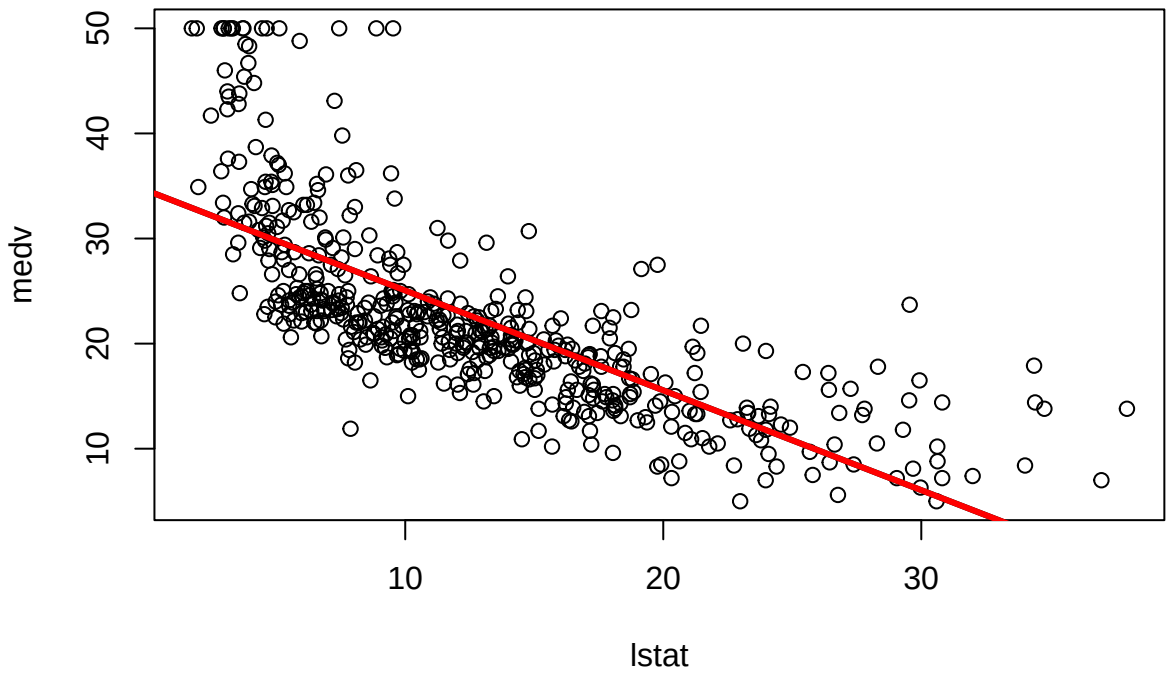
```
plot(lstat, medv)
abline(lm.fit)
```



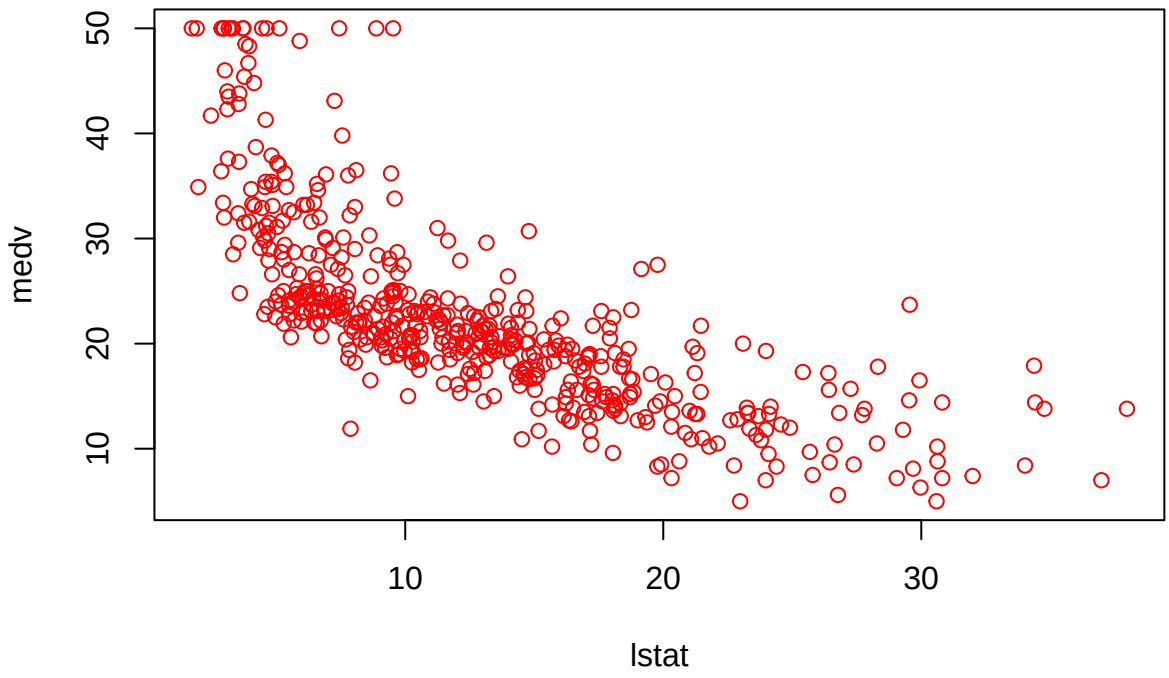
There is some evidence for non-linearity in the relationship between `lstat` and `medv`. We will explore this issue later.

The `abline()` function can be used to draw any line, not just the least squares regression line. To draw a line with intercept `a` and slope `b`, we type `abline(a, b)`. Below we experiment with some additional settings for plotting lines and points. The `lwd = 3` command causes the width of the regression line to be increased by a factor of 3; this works for the `plot()` and `lines()` functions also. We can also use the `pch` option to create different plotting symbols.

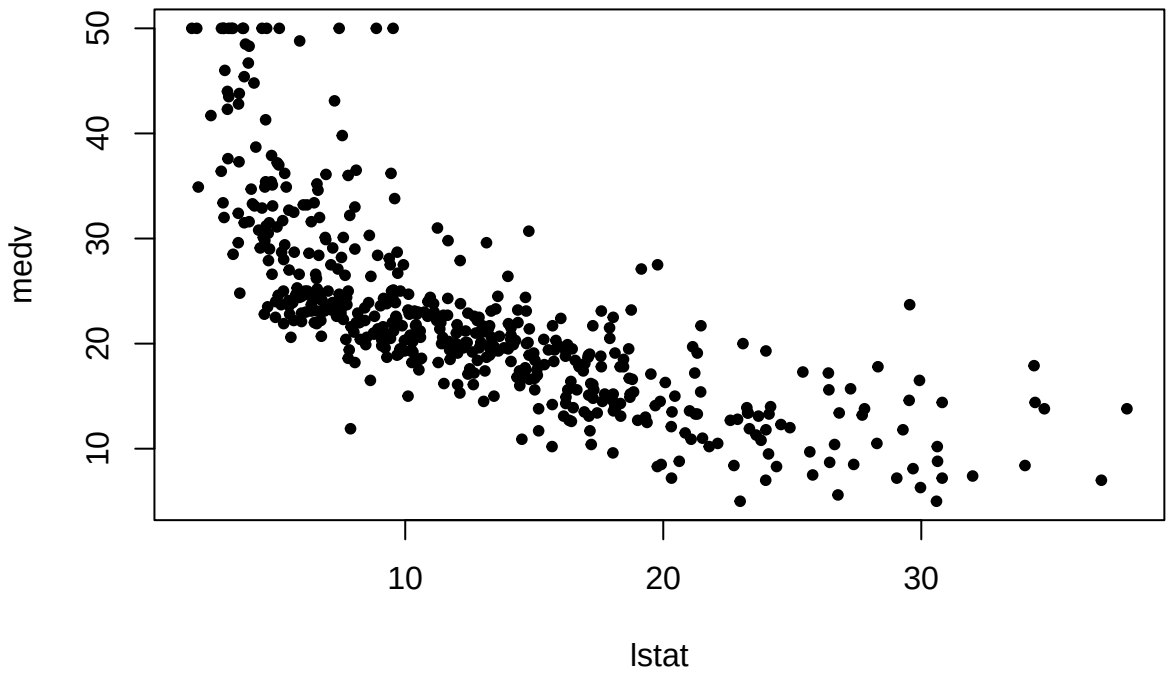
```
plot(lstat, medv)
abline(lm.fit, lwd = 3)
abline(lm.fit, lwd = 3, col = "red")
```



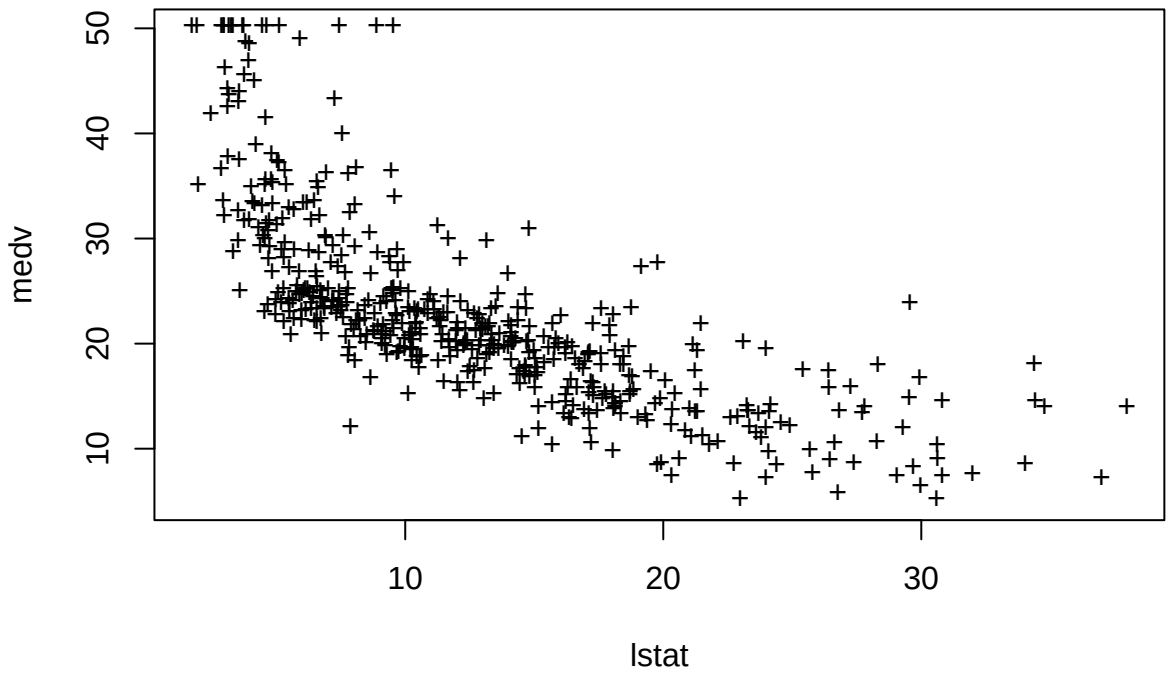

```
plot(lstat, medv, col = "red")
```




```
plot(lstat, medv, pch = 20)
```




```
plot(lstat, medv, pch = "+")
```


medv
10
20
30
40
50
10
20
30
lstat